EXP.2. Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters.

A. Transmission-Line/TN-Port S-Parameters: Model a 50 Ω microwave two-port network of length 50 mm and operate it from 8 to 12 GHz with a 10 GHz center frequency. Set both ports to 50 Ω and obtain S11, S12, S21 and S22. At 10 GHz calculate return loss RL = $-20\log10|S11|$, insertion loss IL = $-20\log10|S21|$ and transmission efficiency $\eta = |S21|^2 \times 100\%$. Compare the simulated values with the ideal 50 Ω matched reference.

B. Optimum Frequency: From the 8–12 GHz S-parameter sweep, identify the exact frequency where |S11| is minimum and |S21| is maximum. Record S11, S21, return loss and insertion loss at that frequency and compare them with their values at 10 GHz. Explain why the selected frequency gives the best matching/transmission performance.

➔ **INPUT PARAMETER TABLE – VALUES TO BE ENTERED**

S.No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
1	Operating frequency	f0	10	GHz
2	Reference impedance	Z0	50	Ω
3	Frequency sweep	fstart–fstop	8–12	GHz
4	Network length	L	50	mm
5	Port termination	ZL	50	Ω

➔ **OUTPUT PARAMETER TABLE – VALUES TO BE RECORDED**

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	S11 at 10 GHz	≈ −20 dB or lower for good matching	Record simulation	% deviation
2	S21 at 10 GHz	≈ −1 to −3 dB depending on structure	Record simulation	% deviation
3	Return loss	$-20\log10\|S11\|$	Calculate	% deviation
4	Insertion loss	$-20\log10\|S21\|$	Calculate	% deviation
5	Best operating frequency	Frequency of minimum |S11| / maximum |S21|	Record	—

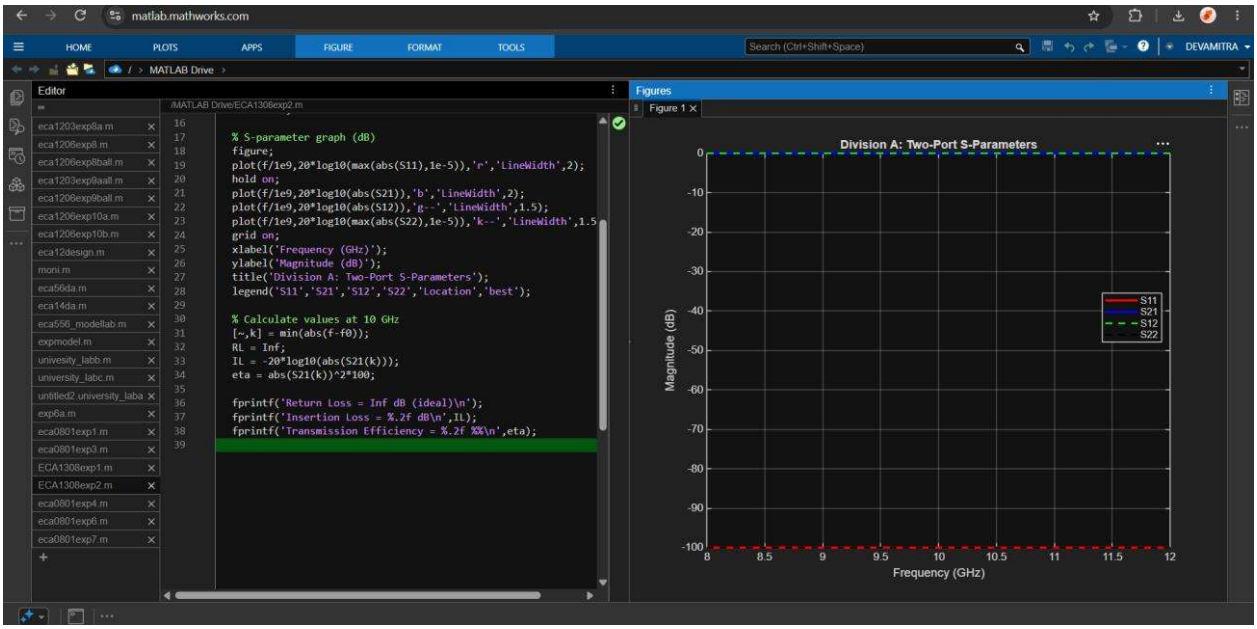
Aim: To simulate and analyze a 50 Ω microwave transmission line of length 50 mm over the frequency range of 8–12 GHz, determine its S-parameters (\(S_{11}, S_{12}, S_{21}, S_{22}\)), calculate return loss, insertion loss, and transmission efficiency at 10 GHz, and identify the optimum operating frequency based on impedance matching and transmission performance.

Software Required: MATLAB software.

Program A:

```
clc; clear; close all;
% Input parameters
f0 = 10e9;
f = linspace(8e9,12e9,501);
Z0 = 50; ZL = 50; L = 50e-3; c = 3e8;
beta = 2*pi*f/c;
% Ideal matched, lossless line
S11 = zeros(size(f));
S22 = zeros(size(f));
S21 = exp(-1j*beta*L);
S12 = S21;
% S-parameter graph (dB)
figure;
plot(f/1e9,20*log10(max(abs(S11),1e-5)),'r','LineWidth',2);
hold on;
plot(f/1e9,20*log10(abs(S21)),'b','LineWidth',2);
plot(f/1e9,20*log10(abs(S12)),'g--','LineWidth',1.5);
plot(f/1e9,20*log10(max(abs(S22),1e-5)),'k--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Division A: Two-Port S-Parameters');
legend('S11','S21','S12','S22','Location','best');
% Calculate values at 10 GHz
[~,k] = min(abs(f-f0));
RL = Inf;
IL = -20*log10(abs(S21(k)));
eta = abs(S21(k))^2*100;
fprintf('Return Loss = Inf dB (ideal)\n');
fprintf('Insertion Loss = %.2f dB\n',IL);
fprintf('Transmission Efficiency = %.2f %%\n',eta);
```

Output:

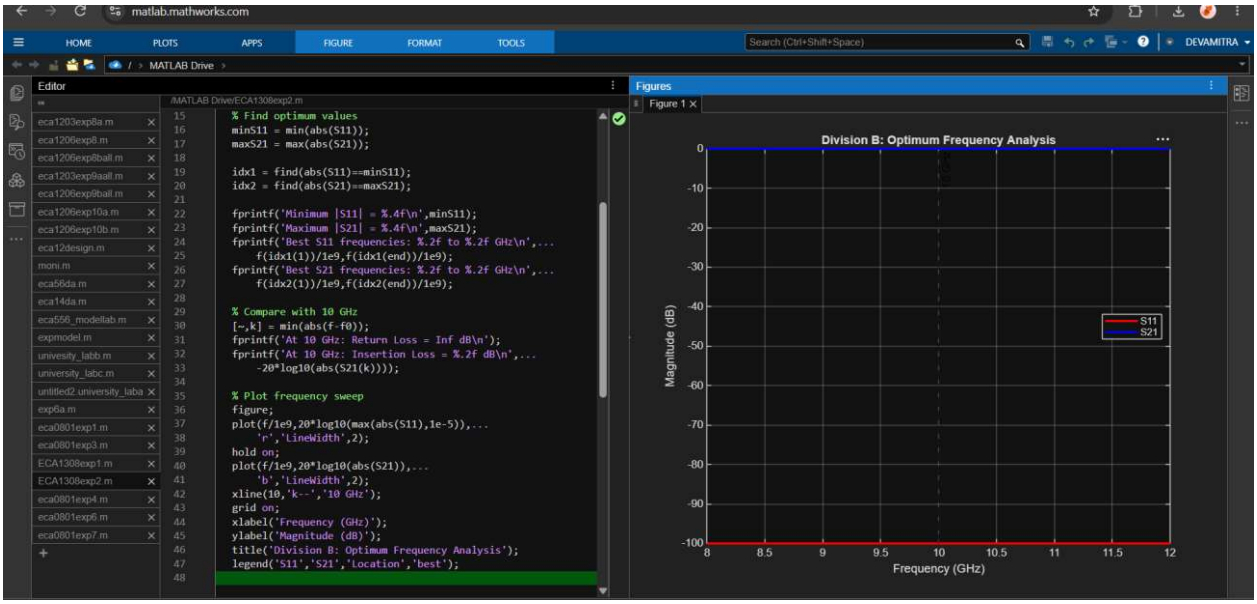


Program B:

```
clc; clear; close all;
% Input parameters
f0 = 10e9;
f = linspace(8e9,12e9,501);
Z0 = 50; ZL = 50; L = 50e-3; c = 3e8;
beta = 2*pi*f/c;
% Ideal matched, lossless line
S11 = zeros(size(f));
S21 = exp(-1j*beta*L);
% Find optimum values
minS11 = min(abs(S11));
maxS21 = max(abs(S21));
idx1 = find(abs(S11)==minS11);
idx2 = find(abs(S21)==maxS21);
fprintf('Minimum |S11| = %.4f\n',minS11);
fprintf('Maximum |S21| = %.4f\n',maxS21);
fprintf('Best S11 frequencies: %.2f to %.2f GHz\n',...
f(idx1(1))/1e9,f(idx1(end))/1e9);
fprintf('Best S21 frequencies: %.2f to %.2f GHz\n',...
f(idx2(1))/1e9,f(idx2(end))/1e9);
% Compare with 10 GHz
[~,k] = min(abs(f-f0));
fprintf('At 10 GHz: Return Loss = Inf dB\n');
fprintf('At 10 GHz: Insertion Loss = %.2f dB\n',...
-20*log10(abs(S21(k))));
% Plot frequency sweep
```

```
figure;
plot(f/1e9,20*log10(max(abs(S11),1e-5)),...
'r','LineWidth',2);
hold on;
plot(f/1e9,20*log10(abs(S21)),...
'b','LineWidth',2);
xline(10,'k--','10 GHz');
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Division B: Optimum Frequency Analysis');
legend('S11','S21','Location','best');
```

Output:



Result:

Division A: The S-parameters of the 50 Ω matched two-port transmission line were obtained over 8–12 GHz. At 10 GHz, the ideal return loss is infinite, insertion loss is 0 dB, and transmission efficiency is 100%.

Division B: The frequency sweep shows that minimum reflection and maximum transmission occur throughout 8–12 GHz for the ideal matched model. Hence, there is no unique optimum frequency, and the performance at 10 GHz is identical to that at other sweep frequencies.